Mohammad Sefid

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RESEARCH PROFILE

Master's-level researcher in reconfigurable intelligent surface (RIS)-aided integrated sensing and communication (ISAC) systems, with a focus on cooperative non-orthogonal multiple access (NOMA) and simultaneous wireless information and power transfer (SWIPT) energy harvesting. Experienced in deep reinforcement learning (DRL), alternating optimization (AO) and metaheuristic methods for wireless resource allocation and beamforming, supported by graduate teaching experience in optimization, estimation and information theory.

RESEARCH EXPERIENCE

Graduate Researcher — RIS-Assisted ISAC Systems — *K. N. Toosi University of Technology, Tehran, Iran*
MSc research, 2023–2026

- Performance improvement of RIS-assisted ISAC systems, jointly addressing communication and sensing performance in next-generation wireless networks.
- Cooperative NOMA with SWIPT energy harvesting, comparing system designs for reliable and energy-efficient operation.
- Optimization using deep reinforcement learning, alternating optimization and metaheuristic techniques, implemented and evaluated in Python and MATLAB.
- Course-linked research projects in machine learning, numerical optimization, digital signal processing and adaptive filters, with code-based deliverables.

PUBLICATIONS

Manuscript in progress

- "IRS-Assisted Cooperative NOMA with SWIPT in ISAC Systems: A Comparison of Half-Duplex and Full-Duplex Modes." Status: in progress — not yet published.

TEACHING EXPERIENCE

Teaching Assistant — *K. N. Toosi University of Technology • Feb 2025 – Jan 2026*
Graduate courses: Numerical Optimization, Stochastic Processes, Estimation Theory and Information Theory

- Reviewed and graded computer-based and theoretical assignments for graduate students.
- Prepared and provided correct code and solution keys for assignment questions.

EDUCATION

MSc, Electrical, Electronics and Communications Engineering — *2023–2026*
K. N. Toosi University of Technology, Tehran, Iran — Thesis research: RIS-aided ISAC with cooperative NOMA and SWIPT
Relevant coursework: Machine Learning, Numerical Optimization, Digital Signal Processing, Adaptive Filters.

BSc, Electrical and Electronics Engineering
University of Applied Science and Technology, Iran — Project: A Review of Photovoltaic (PV) Technologies: System Components, Architectures, and Future Research Directions

Associate Degree, Electrical and Electronics Engineering
Arak University of Technology, Iran

TECHNICAL SKILLS

- **Programming and tools:** Python, MATLAB, GitHub, LaTeX, Microsoft Office
- **Methods:** Deep Reinforcement Learning (DRL), Alternating Optimization (AO), metaheuristics (PSO, GWO)
- **Research areas:** RIS/IRS, ISAC, Cooperative NOMA, SWIPT

CONFERENCE PRESENTATION

- Conference on Electrical and Computer Innovations, K. N. Toosi University of Technology, Feb 2025 — presented research on "Applications of STAR RIS in Communication Systems."

RESEARCH INTERESTS

- Joint optimization of integrated communication and sensing performance in next-generation wireless networks
- SWIPT for cooperative NOMA systems
- Optimization of reconfigurable intelligent surfaces to enhance beamforming of wireless signals
- Deep reinforcement learning for adaptive wireless systems

INDUSTRY EXPERIENCE

Intern — Manasazan Sadid — *Tehran, Iran • Apr – Jun 2021*

- Gained practical experience in the planning and design of renewable energy power plants; supported installation of energy systems, including off-grid solar power plants; developed understanding of installation project management and logistics coordination.

Intern — Zamyad Co. — *Tehran, Iran • Jul – Sep 2018*

- Developed knowledge of automotive electrical and electronic systems; provided electrical and electronic support for quality control laboratories and manufacturing lines; evaluated safety measures and contributed to installation of grounding (earthing) systems for technical lines.

Technical Content Creator — Ariosunshade — *Tehran, Iran • Sep 2017 – Mar 2018*

- Translated technical energy-related articles from English into Persian; conducted literature reviews to identify state-of-the-art advancements in the energy sector.

ADDITIONAL

- Half marathon runner
- Recreational cyclist
- Regular workout routine